

Q1 Find the first four terms of the series whose n^{th} term is given by:

(a) $T_n = 4n + 1$

(b) $T_n = 3n - 6$

(c) $T_n = \frac{4 - 3n}{5}$

Q2 Consider the arithmetic progression 2, 5, 8,

(a) Write down the expression for the general term (i.e. n^{th} term).

(b) Find the 50th term.

(c) Find the sum of the first 30 terms.

Q3 Consider the arithmetic progression 15, 7, -1,

(a) Write down the expression for the general term (i.e. n^{th} term).

(b) Find the 20th term.

(c) The sum of k terms is -6258. Find k .

Q4 Find an expression for the n^{th} term of an arithmetic progression if:

(a) the tenth term is 27 and $T_{17} = 41$.

(b) $T_1 = 25$ and $T_9 = 1$.

(c) $T_5 = 17$ and $S_5 = 55$.

(d) the sum of the first 10 terms is 230 and the sum of the first 20 terms is -140.

(e) the 6th term is 4 times the 3rd and also the sum of the 4th and 5th terms is 15.

Q5 (a) How many terms are in the series $2 + 5 + 8 + 11 + \dots + 299$?

(b) $5 + 7 + 9 + \dots + x = 5772$. Find x .

Q6 The sum of the first n terms of an arithmetic series is given by $S_n = \frac{n(7n-5)}{2}$.

(a) Calculate S_1 and S_2 .

(b) Find the first 3 terms of this series.

(c) Find an expression for the n^{th} term.

Q7 The sum of the first n terms of an arithmetic series is given by $S_n = n(11 - 7n)$.

(a) Calculate S_1 and S_2 .

(b) Find the first 3 terms of this series.

(c) Find an expression for the n^{th} term.

1(a) $T_1 = 4(1) + 1 = 5$ $T_2 = 4(2) + 1 = 9$ $T_3 = 4(3) + 1 = 13$	3(c) <i>cont.....</i> Using quadratic formula: $k = \frac{19 \pm \sqrt{(-19)^2 - 4(4)(-6258)}}{2(4)} = \frac{19 \pm 317}{8}$ $= 42, -37\frac{1}{4}$ $\therefore k = 42$ ($k > 0$ and k is an integer)
1(b) $T_1 = 3(1) - 6 = -3$ $T_2 = 3(2) - 6 = 0$ $T_3 = 3(3) - 6 = 3$	4(a) $T_{10} = 27 \Rightarrow a + 9d = 27$(1) $T_{17} = 41 \Rightarrow a + 16d = 41$(2) $(2) - (1) \Rightarrow \therefore 7d = 14$ $\therefore d = 2$ sub into (1) $\Rightarrow a + 9(2) = 27$ $\therefore a = 9$ $\therefore T_n = 9 + (n-1) \cdot 2$ $= 9 + 2n - 2$ $= 2n + 7$
1(c) $T_1 = \frac{4-3(1)}{5} = \frac{1}{5}$ $T_2 = \frac{4-3(2)}{5} = -\frac{2}{5}$ $T_3 = \frac{4-3(3)}{5} = -1$	4(b) $T_1 = 25 \Rightarrow a = 25$ $T_9 = 1 \Rightarrow a + 8d = 1$ sub $a = 25 \Rightarrow 25 + 8d = 1$ $\therefore 8d = -24$ $\therefore d = -3$ $\therefore T_n = 25 + (n-1) \cdot (-3)$ $= 25 - 3n + 3$ $= 28 - 3n$
2(a) 2, 5, 8, A.P. $a = 2, d = 3$ $T_n = a + (n-1)d$ $\therefore T_n = 2 + (n-1) \cdot 3$ $= 2 + 3n - 3$ $= 3n - 1$	4(c) $T_5 = 17 \Rightarrow a + 4d = 17$(1) $S_5 = 55 \Rightarrow \frac{5}{2}(2a + 4d) = 55$ $\therefore 5a + 10d = 55$ $(\div 5) \therefore a + 2d = 11$(2) $(1) - (2) \Rightarrow 2d = 6$ $\therefore d = 3$ sub into (1) $\Rightarrow a + 4(3) = 17$ $\therefore a = 5$ $\therefore T_n = 5 + (n-1) \cdot 3$ $= 5 + 3n - 3$ $= 3n + 2$
2(b) $T_{50} = 3(50) - 1$ $= 149$	
2(c) $S_n = \frac{n}{2}(2a + (n-1)d)$ $\therefore S_{30} = \frac{30}{2}(2(2) + 29 \times 3)$ $= 1365$	
3(a) 15, 7, -1, A.P. $a = 15, d = -8$ $\therefore T_n = 15 + (n-1) \cdot (-8)$ $= 15 + 8n - 8$ $= 23 - 8n$	
3(b) $T_{20} = 23 - 8(20)$ $= -137$	
3(c) $S_k = -6258$ $\therefore \frac{k}{2}[2(15) + (k-1) \cdot (-8)] = -6258$ $\therefore \frac{k}{2}(30 - 8k + 8) = -6258$ $\therefore \frac{k}{2}(38 - 8k) = -6258$ $\therefore 19k - 4k^2 = -6258$ $\therefore 4k^2 - 19k - 6258 = 0$	

4(d) $S_{10} = 230 \Rightarrow \frac{10}{2}(2a + 9d) = 230$ $(\div 5) \therefore 2a + 9d = 46 \dots\dots(1)$ $S_{20} = -140 \Rightarrow \frac{20}{2}(2a + 19d) = -140$ $(\div 10) \therefore 2a + 19d = -14 \dots\dots(2)$ $(2) - (1) \Rightarrow 10d = -60$ $\therefore d = -6$ sub into (1) $\Rightarrow 2a + 9(-6) = 46$ $\therefore a = 50$ $\therefore T_n = 50 + (n-1)(-6)$ $= 56 - 6n$	5(b) $5 + 7 + 9 + \dots x = 5772$ A.P. $a = 5, d = 2, n = ???$ $S_n = 5772$ $\therefore \frac{n}{2}(2(5) + (n-1) \cdot 2) = 5772$ $\therefore \frac{n}{2}(10 + 2n - 2) = 5772$ $\therefore 5n + n^2 - n = 5772$ $\therefore n^2 + 4n - 5772 = 0$ Using quadratic formula: $n = \frac{-4 \pm \sqrt{4^2 - 4(1)(-5772)}}{2(1)} = \frac{-4 \pm 152}{2}$ $\therefore n = 74$ (ignoring $n < 0$) Now $x = T_{74} = 5 + (73) \cdot 2 = 151$
4(e) $T_6 = 4T_3$ $\therefore a + 5d = 4(a + 2d)$ $\therefore a + 5d = 4a + 8d$ $\therefore 3a + 3d = 0$ $(\div 3) \therefore a + d = 0 \dots\dots(1)$ Also $T_4 + T_5 = 15$ $\therefore (a + 3d) + (a + 4d) = 15$ $\therefore 2a + 7d = 15 \dots\dots(2)$ $(1) \times (2) \Rightarrow 2a + 2d = 0 \dots\dots(3)$ $(2) - (3) \Rightarrow \therefore 5d = 15$ $\therefore d = 3$ sub into (1) $\Rightarrow a + 3 = 0$ $\therefore a = -3$ $\therefore T_n = -3 + (n-1) \cdot 3$ $= -3 + 3n - 3$ $= 3n - 6$	6(a) $S_n = \frac{n(7n-5)}{2}$ $\therefore S_1 = \frac{1(7(1)-5)}{2} = 1$ $\therefore S_2 = \frac{2(7(2)-5)}{2} = 9$ 6(b) $S_1 = 1 \Rightarrow 1, \dots, \dots, \dots,$ $S_2 = 9 \Rightarrow 1, 8, \dots, \dots,$ $\therefore$ A.P. is $1, 8, 15, \dots$ where $a = 1, d = 7$ 6(c) $\therefore T_n = 1 + (n-1) \cdot 7$ $= 1 + 7n - 7$ $= 7n - 6$
5(a) $2 + 5 + 8 + \dots + 299$ A.P. where $a = 2, d = 3, n = ???$ $T_n = 299$ $\therefore 2 + (n-1)3 = 299$ $\therefore 3n - 3 = 297$ $\therefore 3n = 300$ $\therefore n = 100$ $\therefore$ There are 100 terms.	7(a) $S_n = n(11 - 7n)$ $\therefore S_1 = 1(11 - 7(1)) = 4$ $\therefore S_2 = 2(11 - 7(2)) = -6$ 7(b) $S_1 = 4 \Rightarrow 4, \dots, \dots, \dots,$ $S_2 = -6 \Rightarrow 4, -10, \dots, \dots,$ $\therefore$ A.P. is $4, -10, -24, \dots$ where $a = 4, d = -14$ 7(c) $\therefore T_n = 4 + (n-1)(-14)$ $= 4 - 14n + 14$ $= 18 - 14n$

Q1 Find the first three terms of the series whose n^{th} term is given by:

(i) $T_n = 2^n$ (ii) $T_n = 8\left(\frac{3}{2}\right)^n$ (iii) $T_n = 64 \times 2^{1-2n}$

Q2 Consider the geometric progression 2, 6, 18,

- (i) Write down the expression for the general term (i.e. n^{th} term).
- (ii) Find the 15th term.
- (iii) Find the sum of the first 10 terms.

Q3 Consider the geometric progression 1024, -768, 576, -432,

- (i) Write down the expression for the general term (i.e. n^{th} term).
- (ii) Find the 12th term. Answer in the form $\frac{a}{b}$ where a and b are integers.
- (iii) Find the sum of the first 8 terms.

Q4 Find an expression for the n^{th} term of a geometric progression if:

- (i) the third term is 5 and the ratio of the 6th term to the 5th term is -2.
- (ii) $T_1 = 32$ and $T_4 = 108$.
- (iii) $T_4 = 10$ and $T_{10} = 640$.

Q5 (i) How many terms are in the sequence $\frac{1}{4}, \frac{1}{2}, 1, 2, \dots, 4096$?

- (ii) $2 + 6 + 18 + \dots + x = 531\,440$. Find x .

ANSWERS

- A) 16
- B) 2,4,8
- C) 526.5625
- D) 14
- E) 12,18,27
- F) 64,32,8
- G) 9,565,938
- H) 15
- I) 354,296
- J) 59,036
- K) 525.5625
- L) 9,565,944
- M) 59,048
- N) 8,12,18
- O) 32,8,2
- P) 526.5265
- Q) 354,294
- R) $\frac{5}{4}(-2)^{n-1}$
- S) $1,024\left(\frac{3}{4}\right)^{n-1}$
- T) $\frac{3}{4}(2)^{n-1}, -\frac{3}{4}(2)^{n-1}$
- U) $2.(3)^{n-1}$
- V) $-\frac{177,147}{4096}$
- W) $\frac{5}{4}(2)^{n-1}, -\frac{5}{4}(2)^{n-1}$
- X) $1,024\left(\frac{-3}{4}\right)^{n-1}$
- Y) $32.(1.5)^{n-1}$
- Z) $\frac{3}{4}(-2)^{n-1}$

Geometric Progressions - Solutions

1(i) $T_1 = 2^1 = 2$ $T_2 = 2^2 = 4$ $T_3 = 2^3 = 8$	3(iii) $S_8 = \frac{1024\left(1 - \left(-\frac{3}{4}\right)^8\right)}{1 - \left(-\frac{3}{4}\right)}$ $= 526.5625$
1(ii) $T_1 = 8\left(\frac{3}{2}\right)^1 = 12$ $T_2 = 8\left(\frac{3}{2}\right)^2 = 18$ $T_3 = 8\left(\frac{3}{2}\right)^3 = 27$	Q4 $T_n = ar^{n-1}$ 4(i) $T_3 = 5, \frac{T_6}{T_5} = -2 \Rightarrow \therefore r = -2$ $T_3: a(-2)^{3-1} = 5$ $\therefore 4a = 5$ $\therefore a = \frac{5}{4}$ $\therefore T_n = \frac{5}{4}(-2)^{n-1}$
1(iii) $T_1 = 64 \times 2^{1-2} = 32$ $T_2 = 64 \times 2^{1-4} = 8$ $T_3 = 64 \times 2^{1-6} = 2$	
Q2 2, 6, 18, ... is a G.P. $a = 2, r = 3$ 2(i) $T_n = ar^{n-1}$ $\therefore T_n = 2 \times 3^{n-1}$ 2(ii) $T_{15} = 2 \times 3^{15-1}$ $= 9\,565\,938$ 2(iii) $S_n = \frac{a(1-r^n)}{1-r}$ $\therefore S_{10} = \frac{2(1-3^{10})}{1-3}$ $= 59\,048$	4(ii) $T_1 = 32 \Rightarrow \therefore a = 32$ $T_4 = 108$ $T_4: 32r^{4-1} = 108$ $\therefore r^3 = \frac{108}{32}$ $\sqrt[3]{} \therefore r = 1.5$ $\therefore T_n = 32(1.5)^{n-1}$
Q3 1024, -768, 576, ... is a G.P. $a = 1024, r = -\frac{3}{4}$ 3(i) $T_n = 1024\left(-\frac{3}{4}\right)^{n-1}$ 3(ii) $T_{12} = 1024\left(-\frac{3}{4}\right)^{12-1}$ $= 2^{10} \times \frac{(-3)^{11}}{(2^2)^{11}}$ $= \frac{2^{10} \times (-3)^{11}}{2^{22}}$ $= \frac{(-3)^{11}}{2^{12}}$ $= -\frac{177\,147}{4096}$	4(iii) $T_4 = 10 \Rightarrow ar^3 = 10$ $T_{10} = 640 \Rightarrow ar^9 = 640$ By division $\therefore r^6 = 64$ $\therefore r = \pm 2$ when $r = 2, a(2)^3 = 10$ $\therefore 8a = 10$ $\therefore a = \frac{5}{4}$ when $r = -2, a(-2)^3 = 10$ $\therefore -8a = 10$ $\therefore a = -\frac{5}{4}$ $\therefore T_n = \frac{5}{4}(2)^{n-1}$ or $\therefore T_n = -\frac{5}{4}(2)^{n-1}$

5(i) $\frac{1}{4}, \frac{1}{2}, 1, 2, \dots, 4096$ (G.P.)

where $a = \frac{1}{4}, r = 2, n = ???$

$$T_n \Rightarrow ar^{n-1} = 4096$$

$$\therefore \frac{1}{4}(2)^{n-1} = 4096$$

$$\text{using calculator} \quad \therefore 2^{n-1} = 16384$$

$$16384 = 2^{14} \quad \therefore 2^{n-1} = 2^{14}$$

$$\therefore n-1 = 14$$

$$\therefore n = 15$$

$\therefore$ There are 15 terms.

5(ii) $2 + 6 + 18 + \dots + x = 531440$

This is a G.P., $a = 2, r = 3, n = ??$

$$\text{Now } S_n = \frac{a(1-r^n)}{1-r} = 531440$$

$$\therefore \frac{2(1-3^n)}{1-3} = 531440$$

$$\therefore \frac{2(1-3^n)}{-2} = 531440$$

$$\therefore 3^n - 1 = 531440$$

$$\therefore 3^n = 531441$$

$$\text{i.e. } 3^n = 3^{12}$$

$$\therefore n = 12$$

$$\text{using calculator} \quad 531441 = 3^{12}$$

Now $x = T_{12}$ (the 12th term)

$$= 2(3)^{12-1}$$

$$= 354294$$

Q1 Evaluate:

$$(i) \sum_{1}^{12} (3n - 4)$$

$$(ii) \sum_{1}^{20} (24 - 5n)$$

$$(iii) \sum_{3}^{10} (6n - 5)$$

$$(iv) \sum_{1}^{8} 2^n$$

$$(v) \sum_{1}^{10} 2 \times 4^{n-2}$$

$$(vi) \sum_{5}^{12} 64 \left(\frac{1}{2}\right)^{n-4}$$

Q2 Evaluate:

$$(i) \sum_{1}^{\infty} 6 \left(\frac{1}{4}\right)^n$$

$$(ii) \sum_{1}^{\infty} 5 \left(\frac{3}{4}\right)^{n-1}$$

$$(iii) \sum_{1}^{\infty} 2^{4-n}$$

Q3 Evaluate:

$$(i) \sum_{1}^{10} (3n + 3^n) \longrightarrow \text{NOTE: } \sum_{1}^{10} (3n + 3^n) = \sum_{1}^{10} 3n + \sum_{1}^{10} 3^n$$

$$(ii) \sum_{1}^7 \left[\left(\frac{1}{2}\right)^{3-n} - 5n \right]$$

$$(iii) \sum_{4}^{10} (3 \times 6^{n-5} + 2)$$

ANSWERS

- A) 188
- B) 272
- C) -108.25
- D) 268
- E) -570
- F) -510
- G) 28,007.5
- H) 88,737
- I) 174,762
- J) 186
- K) 570
- L) 64
- M) 16
- N) 125
- O) 510
- P) 54
- Q) 4
- R) 2
- S) 88,732
- T) 20
- U) 108.25
- V) 28,007
- W) 63.75
- X) -268
- Y) 15
- Z) 174,762.5

Sigma (Summation) Notation - Solutions

$1(i) \sum_{n=1}^{12} (3n-4) = -1+2+5+\dots+32 \quad (\text{A.P.})$ $= \frac{n}{2}(a+l) \text{ where } n=12, a=-1, l=32$ $= \frac{12}{2}(-1+32)$ $= 186$	$2(i) \sum_{n=1}^{\infty} 6\left(\frac{1}{4}\right)^n = 1\frac{1}{2} + \frac{3}{8} + \frac{3}{32} + \dots \quad \text{LIMITING SUM } \frac{a}{1-r}$ $= \frac{1\frac{1}{2}}{1-\frac{1}{4}}$ $= 2$
$1(ii) \sum_{n=1}^{20} (24-5n) = 19+14+9+\dots-76 \quad (\text{A.P.})$ $= \frac{n}{2}(a+l) \text{ where } n=20, a=19, l=-76$ $= \frac{20}{2}(19-76)$ $= -570$	$2(ii) \sum_{n=1}^{\infty} 5\left(\frac{3}{4}\right)^{n-1} = 5+3\frac{3}{4}+2\frac{13}{16}+\dots \quad \text{LIMITING SUM } \frac{a}{1-r}$ $= \frac{5}{1-\frac{3}{4}}$ $= 20$
$1(iii) \sum_{n=3}^{10} (6n-5)$ $= 13+19+25+\dots+55 \quad (\text{A.P.})$ $= \frac{n}{2}(a+l) \text{ where } n=8, a=13, l=55$ $= \frac{8}{2}(13+55)$ $= 272$	$2(iii) \sum_{n=1}^{\infty} 2^{4-n} = 8+4+2+\dots \quad \text{LIMITING SUM } \frac{a}{1-r}$ $= \frac{8}{1-\frac{1}{2}}$ $= 16$
$1(iv) \sum_{n=1}^8 2^n = 2+4+8+\dots \quad (\text{G.P.})$ $= \frac{a(1-r^n)}{1-r} \text{ where } a=2, r=2, n=8$ $= \frac{2(1-2^8)}{1-2}$ $= 510$	$3(i) \sum_{n=1}^{10} 3n+3^n = \sum_{n=1}^{10} 3n + \sum_{n=1}^{10} 3^n$ $= (3+6+9+\dots+30) + (3+9+27+\dots)$ $= \frac{10}{2}(3+30) + \frac{3(1-3^{10})}{1-3}$ $= 88737 \quad \text{A.P.} \quad \text{G.P.}$
$1(v) \sum_{n=1}^{10} 2 \times 4^{n-2} = 0.5+2+8+\dots \quad (\text{G.P.})$ $= \frac{a(1-r^n)}{1-r} \text{ where } a=0.05, r=4, n=10$ $= \frac{0.5(1-4^{10})}{1-4}$ $= 174762.5$	$3(ii) \sum_{n=1}^7 \left[\left(\frac{1}{2}\right)^{3-n} - 5n \right] = \sum_{n=1}^7 \left(\frac{1}{2}\right)^{3-n} - \sum_{n=1}^7 5n$ $= (0.25+0.5+1+\dots) - (5+10+15+\dots+35)$ $= \frac{0.25(1-2^7)}{1-2} - \frac{7}{2}(5+35)$ $= -108.25 \quad \text{G.P.} \quad \text{A.P.}$
$1(vi) \sum_{n=5}^{12} 64\left(\frac{1}{2}\right)^{n-4} = 32+16+8+\dots \quad (\text{G.P.})$ $= \frac{a(1-r^n)}{1-r} \text{ where } a=32, r=0.5, n=8$ $= \frac{32(1-0.5^8)}{1-0.5}$ $= 63.75$	$3(iii) \sum_{n=4}^{10} (3 \times 6^{n-5} + 2) = \sum_{n=4}^{10} 3 \times 6^{n-5} + \sum_{n=4}^{10} 2$ $= (0.5+3+18+\dots) + (2+2+2+2+\dots+2)$ $= \frac{0.5(1-6^7)}{1-6} + (7 \times 2)$ $= 28007.5 \quad \text{G.P.} \quad \text{A.P.}$

- Q1** (i) The first 3 terms of an arithmetic progression are $2x$, $4x$, $x+15$. Find x .
 (ii) The first 3 terms of an arithmetic progression are $2x-11$, 11 , $2x+1$. Find x .
- Q2** (i) The first 3 terms of a geometric progression are $x-3$, x , $2x$. Find x .
 (ii) The first 3 terms of a geometric progression are $2x+8$, x , $5-x$. Find x .
- Q3** Using limiting sums, express the following as fractions in their simplest form:
 (i) $0.\dot{7}$ (ii) $0.\dot{6}\dot{2}$ (iii) $0.5\dot{3}\dot{1}$
- Q4** The positive multiples of 6 are 6, 12, 18,
 (i) What is the first multiple of 6 greater than 500?
 (ii) Find the sum of all multiples of 6 that are greater than 500 but less than 1000.
- Q5** Peter is employed on a first year salary of \$28 900.
 At the start of each year after the first, his salary increases by \$1 100.
 (i) What would his salary be for his sixth year?
 (ii) If he is able to save 10% of his earnings each year, what are his total savings for the first 10 years?
- Q6** Logs are stacked in a pile such that there are 8 logs on the top row, 9 on the next, 10 on the next, and so on. In total there are 378 logs.
 (i) Find the number of rows.
 (ii) How many logs are in the bottom row?
- Q7** Bricks are stacked in a pile such that there are 7 bricks on the top row, 9 on the next, 11 on the next, and so on. There are n rows altogether.
 (i) Write down the number of bricks in the bottom row (i.e. the n^{th} row).
 (ii) Show that there are $6n + n^2$ bricks altogether. **NOTE: Questions 7(ii), 8(i) and 8(ii) are self marked.**
- Q8** (i) If x , y , z are consecutive terms in a geometric progression, show that $y = \pm\sqrt{xz}$.
 (ii) If x , y , z are consecutive terms in an arithmetic progression, show that $y = \frac{x+z}{2}$.

ANSWERS

- A) 8
 B) $2n+9$
 C) \$33,850
 D) \$33,800
 E) 3
 F) 21
 G) 7
 H) 62,250
 I) 30
 J) 5
 K) 6
 L) 28
 M) 496
 N) \$34,400
 O) 504
 P) $2n-3$
 Q) $2n+5$
 R) 62,500
 S) 29
 T) $\frac{263}{495}$
 U) $\frac{263}{497}$
 V) $-\frac{10}{3}, 4$
 W) $\frac{267}{505}$
 X) $-4, \frac{10}{3}$
 Y) $\frac{7}{9}$
 Z) $\frac{62}{99}$

Mixed Problems - Solutions

1(i) For $2x, 4x, x+15$ to be an A.P. $4x - 2x = x + 15 - 4x$ $\therefore 2x = 15 - 3x$ $\therefore 5x = 15$ $\therefore x = 3$	3(ii) $0.626262... = \frac{62}{100} + \frac{62}{10000} + \frac{62}{1000000} + ...$ $\therefore 0.62 = \frac{a}{1-r}, \text{ where } a = \frac{62}{100}, r = \frac{1}{100}$ $= \frac{\frac{62}{100}}{1 - \frac{1}{100}}$ $= \frac{62}{99}$
1(ii) For $2x-11, 11, 2x+1$ to be an A.P. $11 - (2x - 11) = 2x + 1 - 11$ $\therefore 22 - 2x = 2x - 10$ $\therefore -4x = -32$ $\therefore x = 8$	3(iii) $0.531 = 0.5313131....$ $\therefore 0.531 = 0.5 + \frac{31}{1000} + \frac{31}{100000} + ...$ $= \frac{1}{2} + \frac{a}{1-r}, \text{ where } a = \frac{31}{1000}, r = \frac{1}{100}$ $= \frac{1}{2} + \frac{\frac{31}{1000}}{1 - \frac{1}{100}}$ $= \frac{263}{495}$
2(i) For $x-3, x, 2x$ to be a G.P. $\frac{x}{x-3} = \frac{2x}{x}$ $\therefore \frac{x}{x-3} = 2 \text{ (after cancelling } x\text{'s, since } x \neq 0\text{)}$ $\therefore x = 2x - 6$ $\therefore -x = -6$ $\therefore x = 6$	Q4 $6, 12, 18, \Rightarrow$ A.P. $a = 6, d = 6$ 4(i) $T_n > 500$ $\therefore a + (n-1)d > 500$ $\therefore 6 + (n-1)6 > 500$ $\therefore 6 + 6n - 6 > 500$ $\therefore 6n > 500$ $\therefore n > 83\frac{1}{3} \Rightarrow \therefore n = 84$ Now $T_{84} = 6 + 83(6)$ $= 504$
2(ii) For $2x+8, x, 5-x$ to be a G.P. $\frac{x}{2x+8} = \frac{5-x}{x}$ $\therefore x^2 = (2x+8)(5-x)$ $\therefore x^2 = 10x - 2x^2 + 40 - 8x$ $\therefore 3x^2 - 2x - 40 = 0$ $\therefore (3x+10)(x-4) = 0$ $\therefore x = -\frac{10}{3}, 4$	4(ii) $T_n < 1000$ $\therefore 6 + (n-1)6 < 1000$ $\therefore 6n < 1000$ $\therefore n < 166\frac{2}{3} \Rightarrow \therefore n = 166$ Now $T_{166} = 6 + 165(6) = 996$ $\therefore \text{sum} = 504 + 510 + + 996$ $= \frac{n}{2}(a+l) \text{ where } n = 83, a = 504, l = 996$ $= \frac{83}{2}(504 + 996)$ $= 62250$
3(i) $0.7 = 0.777777.....$ $= \frac{7}{10} + \frac{7}{100} + \frac{7}{1000} + ...$ $\therefore 0.7 = \frac{a}{1-r}, \text{ where } a = \frac{7}{10}, r = \frac{1}{10}$ $= \frac{\frac{7}{10}}{1 - \frac{1}{10}}$ $= \frac{7}{9}$	

Q5 Salary year by year is 28 900, 30 000, 31 100, ... which is an A.P. where $a = 28\,900$, $d = 1100$ 5(i) $T_6 = 28\,900 + (6-1) \cdot 1100$ $= \\$34\,400$ 5(ii) Savings are 2890, 3000, 3110, ... $\therefore S_{10} = \frac{10}{2}(2(2890) + 9 \times 110)$ $= \\$33\,850$	Q7 7, 9, 11, 7(i) $T_n = a + (n-1)d$ $= 7 + (n-1) \cdot 2$ $= 2n + 5$ 7(ii) $S_n = \frac{n}{2}(a+l)$ $= \frac{n}{2}(7 + (2n+5))$ $= \frac{n}{2}(12+2n)$ $= 6n + n^2$ as required
6(i) $8+9+10+\dots = 378$ $S_n = \frac{n}{2}(2a + (n-1)d)$ $\therefore \frac{n}{2}(2(8) + (n-1)1) = 378$ $(\times 2) \therefore n(15+n) = 756$ $\therefore n^2 + 15n - 756 = 0$ $\therefore (n+36)(n-21) = 0$ $\therefore n = 21$ (since $n > 0$) $\therefore$ There are 21 rows.	8(i) For an A.P. $T_2 - T_1 = T_3 - T_2 = \dots$ $\therefore$ For x, y, z to be an A.P. $y - x = z - y$ $\therefore 2y = x + z$ $\therefore y = \frac{x+z}{2}$
6(ii) $T_n = a + (n-1)d$ $\therefore T_{21} = 8 + 20(1)$ $= 28$ $\therefore$ 28 logs on bottom row	8(ii) For a G.P., $\frac{T_2}{T_1} = \frac{T_3}{T_2} = \dots$ $\therefore$ For x, y, z to be a G.P. $\frac{y}{x} = \frac{z}{y}$ $\therefore y^2 = xz$ $\therefore y = \pm\sqrt{xz}$

- Q1** Find the value of each investment after it matures.
(i.e. at the end of the investment period).
- (i) \$5 000 earning 6% p.a. compounded annually for 5 years.
 - (ii) \$8 000 earning 8% p.a. compounded half-yearly for 7 years.
 - (iii) \$3 000 earning 9% p.a. compounded quarterly for 2 years.
 - (iv) \$7 000 earning 9% p.a. compounded monthly for 3 years.

- Q2** \$25 000 invested at $r\%$ p.a. interest compounded annually grows to \$33 386.73 after 4 years.

Find r , the rate of interest.

- Q3** A car initially worth \$28 000 depreciates at the rate of 24% p.a.

- (i) Find out how much the car is worth after 1 year.
- (ii) How much value does the car lose during its 4th year?

- Q4** A block of land overlooking the beach was bought in January 1980.
When it was sold 5 years later, it had doubled in price.

Find the equivalent compound percentage rate p.a. which the price rose over this 5 year period.

ANSWERS

- A) \$13,846.35
- B) \$3,584.49
- C) 7.5
- D) 15.2
- E) \$6,683.77
- F) \$3,575.55
- G) \$9,160.52
- H) 7.8
- I) \$2,875
- J) \$21,280
- K) \$2,950
- L) \$9,240.40
- M) \$6,691.13
- N) \$20,990
- O) 7.4
- P) \$13,853.41
- Q) \$9,200
- R) 14.9

Compound Interest - Solutions

1(i) Balance = $\$5000 \times 1.06^5$ = \$6691.13	Q3 loses 24% of value $\Rightarrow \therefore$ retains 76% p.a.
1(ii) 8% p.a. $\Rightarrow$ 4% per half-year Balance = $\$8000 \times 1.04^{14}$ = \$13853.41	3(i) After 1 year, car's value = $\$28000 \times 0.76$ = \$21280
1(iii) 9% p.a. $\Rightarrow$ 2.25% per quarter Balance = $\$3000 \times 1.0225^8$ = \$3584.49	3(ii) After 3 year, car's value = $\$28000 \times (0.76)^3$ = \$12291.33 $\therefore$ In its 4th year, it loses 24% of \$12291.33 i.e. 4th year loss = 0.24×12291.33 = \$2950 (nearest dollar)
1(iv) 9% p.a. $\Rightarrow$ 0.75% per month 3 years = 36 months Balance = $\$7000 \times 1.0075^{36}$ = \$9160.52	Q4 Let original price be P Let x be the annual increasing factor After 5 years, price has doubled $\therefore P \times x^5 = 2P$ ($\div P$) $\therefore x^5 = 2$ $\sqrt[5]{}$ $\therefore x = 1.148698$ $\therefore$ annual percentage increase = $14.8698\ldots\%$ = 14.9% (1 dec. pl.)
Q2 $25000 \times x^4 = 33386.73$ $\therefore x^4 = \frac{33386.73}{25000}$ $\therefore x^4 = 1.335469\ldots$ $\sqrt[4]{}$ $\therefore x = 1.075$ $\therefore$ rate = $0.075 = 7.5\%$ $\therefore r = 7.5$	

- Q1** An amount of \$2800 is invested in a superannuation fund at the start of each year for 35 years. The fund earns 9% p.a. compounded annually.

How much is the investment worth at the end of that time?

- Q2** An amount of \$800 is invested in a superannuation fund at the start of each year for 45 years. The fund earns 7.25% p.a. compounded annually.

How much is the investment worth at the end of that time?

- Q3** An amount of \$3000 is invested in a superannuation fund at the start of each year for 18 years. The interest at a rate of 9% p.a. is paid every half-year.

How much is the investment worth after 18 years?

- Q4** A man contributes \$1200 each year into a superannuation fund for the first 30 years of his working life. For the next fifteen years (until retirement), he decides to increase this, and invests a total of \$5000 each year.

If the investment earns 8% p.a. paid yearly over the whole year period, how much will his investment be worth upon retiring?

ANSWERS
A) \$138,034
B) \$135,035
C) \$138,134
D) \$138,304
E) \$139,134
F) \$264,241
G) \$264,441
H) \$264,741
I) \$265,341
J) \$612,344
K) \$612,354
L) \$612,444
M) \$613,244
N) \$657,359
O) \$658,349
P) \$658,479

Superannuation - Solutions

Q1 1st \$2800 $\Rightarrow \$2800 \times 1.09^{35}$
2nd \$2800 $\Rightarrow \$2800 \times 1.09^{34}$
↓
last \$2800 $\Rightarrow \$2800 \times 1.09^1$

$\therefore \text{TOTAL} = 2800[1.09 + 1.09^2 + \dots + 1.09^{35}]$
 $= 2800 \times \frac{1.09(1 - 1.09^{35})}{1 - 1.09}$
 $= \$658\,349$ (nearest dollar)

Q2 1st \$800 $\Rightarrow \$800 \times 1.0725^{45}$
2nd \$800 $\Rightarrow \$800 \times 1.0725^{44}$
↓
last \$800 $\Rightarrow \$800 \times 1.0725^1$

$\therefore \text{TOTAL} = 800[1.0725 + 1.0725^2 + \dots + 1.0725^{45}]$
 $= 800 \times \frac{1.0725(1 - 1.0725^{45})}{1 - 1.0725}$
 $= \$264\,241$ (nearest dollar)

Q3 interest rate 9% p.a. $\Rightarrow 4.5\%$ per half-year

1st \$3000 $\Rightarrow \$3000 \times 1.045^{36} \leftarrow 18 \text{ years} = 36 \text{ half-years}$
2nd \$3000 $\Rightarrow \$3000 \times 1.045^{34} \leftarrow 17 \text{ years} = 34 \text{ half-years}$
3rd \$3000 $\Rightarrow \$3000 \times 1.045^{32}$
↓
last \$3000 $\Rightarrow \$3000 \times 1.045^2 \leftarrow 1 \text{ year} = 2 \text{ half-years}$

$\therefore \text{TOTAL} = 3000[1.045^2 + 1.045^4 + \dots + 1.045^{36}]$...which is a sum of G.P. where $r = 1.045^2, n = 18$
 $= 3000 \times \frac{1.045^2(1 - (1.045^2)^{18})}{1 - 1.045^2}$
 $= \$138\,034$ (nearest dollar)

Q4 We will treat this investment in 2 parts:

PART 1	PLUS	PART 2
\$1200 invested each year for the whole period (45 years)		For the last 15 years, the <i>extra</i> \$3800 (to make up the \$5000 to be invested each year for 15 years)
1st \$1200 $\Rightarrow \$1200 \times 1.08^{45}$ 2nd \$1200 $\Rightarrow \$1200 \times 1.08^{44}$ ↓ last \$1200 $\Rightarrow \$1200 \times 1.08^1$		1st \$3800 $\Rightarrow \$3800 \times 1.08^{15}$ 2nd \$3800 $\Rightarrow \$3800 \times 1.08^{14}$ ↓ last \$3800 $\Rightarrow \$3800 \times 1.08^1$
$\therefore \text{TOTAL} = 1200[1.08 + 1.08^2 + \dots + 1.08^{45}]$ $= 1200 \times \frac{1.08(1 - 1.08^{45})}{1 - 1.08}$ $= \$500\,911.28$ $= \$612\,344$ (nearest dollar)	PLUS	$3800[1.08 + 1.08^2 + \dots + 1.08^{15}]$ $+ 3800 \times \frac{1.08(1 - 1.08^{15})}{1 - 1.08}$ $+ \$111\,432.28$

Q1 A loan of \$140 000 is taken over 20 years. The interest rate of 12% p.a. is calculated monthly on the amount owing, with equal monthly repayments, R , to be paid over the life of the loan.

- (i) How much is owed immediately after the first month's repayment has been made?
- (ii) Write an expression for A_n , the amount owing after n months.
- (iii) Calculate the value of R .
- (iv) How much interest is paid over the 20 year period?
- (v) Find the rate of simple interest equivalent to this compound rate.

Q2 Maree borrows \$30 000 at 6% p.a. reducible interest, paying it off in equal monthly instalments.

Calculate the amount of each instalment if she wishes to have the loan paid off after 4 years.

Q3 \$50 000 is lent to two partners in a business. It is to be paid off over two years in *two equal annual* payments, P . Because it is a high risk loan, interest is charged at a rate of 2.5% per month, calculated monthly on the amount owing.

- (i) Find the amount owing after making the first payment.
- (ii) Find P .
- (iii) Find the total interest charged for the loan.

ANSWERS

A) \$704.55	K) \$249,865
B) \$1,540.78	L) $140,000 \times 1.01^n - R(1 + 1.01 + 1.01^2 + \dots)$
C) $140,000 \times 1.01 - R$	M) \$704.55
D) 7.9%	N) 8.5%
E) \$38,567.41	O) $140,000 + 1.01^n - 1.01R$
F) \$705.65	P) \$229,965
G) $140,000 \times 1.01^n - R(1.01 + 1.01^2 + \dots)$	Q) \$27,134.82
H) \$1,541.52	R) 8.2%
I) $140,000 - 1.01R$	S) $50,000 \times 1.025^{12} - P$
J) \$38,657.60	T) $55,000 \times 1.025^{12} - 2P$

Time Payment - Solutions

Q1 12% p.a. $\Rightarrow$ 1% per month

20 years $\Rightarrow$ 240 months

(i) Let A_n be the amount owing after n months

$$\therefore A_1 = 140\,000 \times 1.01 - R$$

$$\begin{aligned} \text{(ii)} \quad A_2 &= A_1 \times 1.01 - R \\ &= (140\,000 \times 1.01 - R) \times 1.01 - R \\ &= 140\,000 \times 1.01^2 - 1.01R - R \\ &= 140\,000 \times 1.01^2 - R(1.01 + 1) \end{aligned}$$

$$\therefore A_3 = 140\,000 \times 1.01^3 - R(1.01^2 + 1.01 + 1)$$

$\downarrow$

$$\therefore A_n = 140\,000 \times 1.01^n - R(1 + 1.01 + 1.01^2 + \dots)$$

n terms

(iii) Loan is paid off after 240 months

$$\text{i.e. } A_{240} = 0$$

$$\therefore 0 = 140\,000 \times 1.01^{240} - R(1 + 1.01 + 1.01^2 + \dots)$$

$$\therefore 140\,000 \times 1.01^{240} = R(1 + 1.01 + 1.01^2 + \dots)$$

$$\therefore R = \frac{140\,000 \times 1.01^{240}}{(1 + 1.01 + 1.01^2 + \dots)}$$

$$\left[\begin{array}{l} \text{sum of a G.P., } a = 1, r = 1.01, n = 240 \\ S_{240} = \frac{1(1 - 1.01^{240})}{1 - 1.01} = 989.255... \end{array} \right]$$

$$\begin{aligned} \therefore R &= \frac{140\,000 \times 1.01^{240}}{989.255...} \\ &= \$1541.52 \end{aligned}$$

(iv) Monthly payments made over 20 years

i.e. over 240 months

$$\begin{aligned} \therefore \text{Total paid back} &= 240 \times \$1541.52 \\ &= \$369\,965 \end{aligned}$$

$$\begin{aligned} \therefore \text{Interest (extra)} &= \$369\,965 - \$140\,000 \\ &= \$229\,965 \end{aligned}$$

(v) $\therefore$ Interest p.a. = $\$229\,965 \div 20$

$$= \$11\,498.25 \text{ (per year)}$$

$$\therefore \text{Interest rate} = \frac{11\,498.25}{140\,000} \times 100 = 8.2\% \text{ p.a.}$$

Q2 6% p.a. $\Rightarrow$ 0.5% per month

to increase by 0.5% $\Rightarrow \times 1.005$

4 years $\Rightarrow$ 48 months

Let A_n be the amount owing after n months

R be the monthly repayment

$$\therefore A_1 = 30\,000 \times 1.005 - R$$

$$\begin{aligned} \therefore A_2 &= A_1 \times 1.01 - R \\ &= (30\,000 \times 1.005 - R) \times 1.005 - R \\ &= 30\,000 \times 1.005^2 - 1.005R - R \\ &= 30\,000 \times 1.005^2 - R(1.005 + 1) \end{aligned}$$

$$\therefore A_3 = 30\,000 \times 1.005^3 - R(1.005^2 + 1.005 + 1)$$

$\downarrow$

$$\therefore A_n = 30\,000 \times 1.005^n - R(1 + 1.005 + 1.005^2 + \dots)$$

n terms

But loan is paid off after 48 months

$$\text{i.e. } A_{48} = 0$$

$$\therefore 0 = 30\,000 \times 1.005^{48} - R(1 + 1.005 + 1.005^2 + \dots)$$

$$\therefore 30\,000 \times 1.005^{48} = R(1 + 1.005 + 1.005^2 + \dots)$$

$$\therefore R = \frac{30\,000 \times 1.005^{48}}{(1 + 1.005 + 1.005^2 + \dots)}$$

$$\left[\begin{array}{l} \text{sum of a G.P., } a = 1, r = 1.005, n = 48 \\ S_{48} = \frac{1(1 - 1.005^{48})}{1 - 1.005} = 54.097... \end{array} \right]$$

$$\begin{aligned} \therefore R &= \frac{30\,000 \times 1.005^{48}}{54.097...} \\ &= \$704.55 \end{aligned}$$

Time Payment - Solutions

3 (i) Now only 2 payments are made, first one after 12 months and the second one after 24 months Let A_n be the amount owing after n months. $\therefore A_1 = 50\,000 \times 1.025$ $\therefore A_2 = A_1 \times 1.025$ $= 50\,000 \times 1.025 \times 1.025$ $= 50\,000 \times 1.025^2$ $\therefore A_3 = A_2 \times 1.025$ $= 50\,000 \times 1.025^2 \times 1.025$ $= 50\,000 \times 1.025^3$ This continues such that the amount owing is increasing by 2.5% <i>per month</i> until the 12th month when a payment, P, is made for the first time. $\therefore A_{12} = 50\,000 \times 1.025^{12} - P$	3 (ii) Now this amount, A_{12}, will increase by 2.5% each month until the second (and final) payment is made after a further 12 months. $\therefore A_{24} = A_{12} \times 1.025^{12} - P$ $= (50\,000 \times 1.025^{12} - P) \times 1.025^{12} - P$ $= 50\,000 \times 1.025^{24} - 1.025^{12} P - P$ $= 50\,000 \times 1.025^{24} - P(1 + 1.025^{12})$ But $A_{24} = 0$ (i.e. loan is paid off) $\therefore 0 = 50\,000 \times 1.025^{24} - P(1 + 1.025^{12})$ $\therefore P = \frac{50\,000 \times 1.025^{24}}{1.025^{12} + 1}$ $= \$38\,567.41$ 3(c) Interest = $(2 \times \\$38\,567.41) - \\$50\,000$ $= \$27\,134.82$
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